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SOFTWARE REQUIREMENTS SPECIFICATION

System Name

Wholesale Business Management & Accounting System

1. FUNCTIONAL REQUIREMENTS

1.1 Authentication & Authorization

FR-001 User Login

System shall allow registered users to login using:

- Email or username
- Password

FR-002 Logout

System shall allow users to logout securely.

FR-003 Password Reset

System shall allow users to reset forgotten passwords.

FR-004 Session Management

System shall automatically expire inactive sessions.

FR-005 Multi-factor Authentication (optional)

System may support OTP/email verification.

1.2 Multi Business Management

FR-006 Create Business

Main admin shall create one or multiple businesses.

Business fields:

- Business Name
- Business Type
- Address
- Phone
- Email
- Currency
- Tax settings

FR-007 Switch Business

User shall switch between owned businesses.

FR-008 Business Isolation

Data of one business shall not mix with another.

1.3 User Management

FR-009 Create Users

Admin shall create users.

Fields:

- Full name
- Email
- Phone
- Username
- Password
- Assigned business
- Role

FR-010 Edit Users

Admin shall edit user details.

FR-011 Delete/Deactivate Users

Admin shall deactivate users.

FR-012 User Activity Logs

System shall track:

- login
 - logout
 - created records
 - deleted records
 - edited records
-

1.4 Roles & Permissions

FR-013 Create Roles

Admin shall create roles.

Examples:

- Cashier
- Accountant
- Sales officer
- Storekeeper

FR-014 Assign Permissions

Admin shall assign permissions per role.

Permissions:

- view
- create
- edit
- delete

- approve

FR-015 Restrict Access

Users shall only access permitted modules.

1.5 Dashboard

FR-016 Dashboard Summary

System shall display:

- Total sales
- Total purchases
- Total receipts
- Total payments
- Total customers
- Total suppliers
- Total stock value
- Low stock alerts

FR-017 Quick Actions

Dashboard shall provide shortcuts:

- create invoice
 - add customer
 - add supplier
 - add receipt
-

1.6 Bank & Cash Accounts

FR-018 Create Account

Admin shall create:

- bank account
- cash account

- mobile wallet

Fields:

- account name
- bank name
- account number
- opening balance

FR-019 Edit Account

System shall allow editing.

FR-020 Delete Account

System shall allow soft delete.

FR-021 Balance Update

Balances shall auto-update after transactions.

1.7 Customers

FR-022 Create Customer

Fields:

- Name
- Credit limit
- Billing address
- Delivery address
- Phone
- Email

FR-023 Edit Customer

FR-024 Delete Customer

FR-025 Customer Search

FR-026 Customer Statement

Generate customer account statement.

FR-027 Customer Balance

Track customer outstanding balance.

1.8 Suppliers

FR-028 Create Supplier

Fields:

- Name
- Credit limit
- Address
- Phone
- Email

FR-029 Edit Supplier

FR-030 Delete Supplier

FR-031 Supplier Search

FR-032 Supplier Statement

FR-033 Supplier Balance

1.9 Inventory Management

FR-034 Create Inventory Item

Fields:

- Item code
- Item name
- Unit name
- Quantity
- Purchase price
- Sales price
- Description

FR-035 Profit Calculation

Profit per item auto-calculated.

FR-036 Edit Inventory

FR-037 Delete Inventory

FR-038 Inventory Search

FR-039 Inventory Filtering

FR-040 Stock Auto Update

Stock decreases after sales.
Stock increases after purchases.

FR-041 Stock Movement History

Track inventory changes.

FR-042 Low Stock Alert

1.10 Receipts

FR-043 Create Receipt

Fields:

- Date
- Paid by

- Received in
- Description

FR-044 Receipt With Items

Receipt may contain sold items.

FR-045 Receipt Without Items

Receipt may record direct cash.

FR-046 Auto Bank Update

Received amount increases bank balance.

FR-047 Receipt CRUD

1.11 Payments

FR-048 Create Payment

Fields:

- Date
- Paid from
- Payee
- Description

FR-049 Payment With Items

FR-050 Payment Without Items

FR-051 Auto Bank Deduction

FR-052 Payment CRUD

1.12 Sales Orders

FR-053 Create Sales Order

Fields:

- Date
- Customer
- Address
- Description

Items:

- Item
- Qty
- Unit price
- Total

FR-054 Edit Sales Order

FR-055 Delete Sales Order

FR-056 Cancel Sales Order

FR-057 Convert to Invoice

1.13 Sales Invoices

FR-058 Create Sales Invoice

FR-059 Edit Sales Invoice

FR-060 Delete Sales Invoice

FR-061 Cancel Sales Invoice

FR-062 Payment Status

Statuses:

- unpaid
- partial
- paid

FR-063 Inventory Reduction

Inventory decreases after invoice.

1.14 Purchase Orders

FR-064 Create Purchase Order

FR-065 Edit Purchase Order

FR-066 Delete Purchase Order

FR-067 Cancel Purchase Order

FR-068 Convert to Purchase Invoice

1.15 Purchase Invoices

FR-069 Create Purchase Invoice

FR-070 Edit Purchase Invoice

FR-071 Delete Purchase Invoice

FR-072 Cancel Purchase Invoice

FR-073 Inventory Increase

Inventory increases after purchase.

1.16 Employees

FR-074 Create Employee

Fields:

- Name
- Phone
- Email
- Address
- Salary
- Position

FR-075 Edit Employee

FR-076 Delete Employee

1.17 Reports

FR-077 Financial Reports

Generate:

- Profit & Loss
- Balance Sheet
- Cash Flow

FR-078 Sales Reports

Generate:

- daily sales
- monthly sales
- yearly sales

FR-079 Purchase Reports

FR-080 Inventory Reports

FR-081 Customer Reports

FR-082 Supplier Reports

FR-083 Export Reports

Export:

- PDF
 - Excel
 - CSV
-

1.18 Search & Filters

FR-084 Global Search

FR-085 Advanced Filters

Filter by:

- date
 - customer
 - supplier
 - status
 - account
-

1.19 CRUD Support

FR-086 Create button

FR-087 Create and Add Another

FR-088 Edit

FR-089 Delete

1.20 Notifications

FR-090 Low Stock Notification

FR-091 Overdue Invoice Notification

FR-092 Due Payment Notification

1.21 Settings

FR-093 Business Settings

FR-094 Tax Settings

FR-095 Currency Settings

FR-096 User Settings

1.22 UI Features

FR-097 Money Blur

All money values blurred until hover.

FR-098 Responsive Layout

FR-099 Sidebar Role Visibility

1.23 Audit & Logs

FR-100 Audit Trail

Track:

- who created
 - who edited
 - who deleted
-
-

2. NON-FUNCTIONAL REQUIREMENTS

Non-functional ni quality attributes.

2.1 Performance

NFR-001 Response Time

Pages load under 3 seconds.

NFR-002 Concurrent Users

Support minimum 500 concurrent users.

NFR-003 Large Dataset Handling

System shall paginate records.

2.2 Security

NFR-004 Password Encryption

Passwords hashed.

NFR-005 HTTPS

All traffic encrypted.

NFR-006 Access Control

Unauthorized users blocked.

NFR-007 SQL Injection Prevention

NFR-008 XSS Protection

NFR-009 CSRF Protection

2.3 Reliability

NFR-010 Availability

99.9% uptime.

NFR-011 Backup

Automatic daily backups.

NFR-012 Recovery

Restore from backup.

2.4 Scalability

NFR-013 Horizontal Scalability

NFR-014 Vertical Scalability

2.5 Maintainability

NFR-015 Clean Architecture

NFR-016 Modular Code

NFR-017 Documentation

2.6 Usability

NFR-018 Easy Navigation

NFR-019 Consistent UI

NFR-020 Minimal Clicks

2.7 Compatibility

NFR-021 Browser Support

- Chrome
- Firefox
- Safari
- Edge

NFR-022 Mobile Support

2.8 Database Integrity

NFR-023 ACID Compliance

NFR-024 Referential Integrity

2.9 Logging

NFR-025 Error Logs

NFR-026 Activity Logs

2.10 Localization

NFR-027 Currency Format

NFR-028 Timezone Support

2.11 Portability

NFR-029 Cloud Deployment

NFR-030 Docker Support

2.12 Accessibility

NFR-031 Keyboard Navigation

NFR-032 Screen Reader Support

2.13 Data Retention

NFR-033 Soft Delete

NFR-034 Archive Old Data

2.14 Compliance

NFR-035 Financial Record Compliance

NFR-036 Tax Compliance

3. BUSINESS RULES

Very important.

BR-001

Inventory cannot go below zero unless negative stock enabled.

BR-002

Deleted invoices must not destroy accounting history.

BR-003

Cancelled invoices restore stock.

BR-004

Receipt increases bank balance.

BR-005

Payment decreases bank balance.

BR-006

One transaction belongs to one business only.

BR-007

Users only access assigned business.

BR-008

Invoices totals auto-calculated.

BR-009

Profit auto-calculated.

BR-010

Customer cannot exceed credit limit unless overridden.
